

Klausurvorbereitung A1

-1-

a) Einzelschrittverfahren: Gauß-Seidel-Verf.

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j < i} a_{ij} x_j^{(k+1)} - \sum_{j > i} a_{ij} x_j^{(k)} \right)$$

$$\underline{k=0} \quad i=1 \quad x_1^{(1)} = \frac{1}{a_{11}} \left(b_1 - 0 - \underbrace{(a_{12} x_2^{(0)} + a_{13} x_3^{(0)})}_{=0} \right)$$

$$= \frac{2}{4} = \underline{\underline{\frac{1}{2}}}$$

$$i=2 \quad x_2^{(1)} = \frac{1}{a_{22}} \left(b_2 - a_{21} x_1^{(1)} - a_{23} x_3^{(0)} \right)$$

$$= \frac{1}{4} \left(0 - 2 \cdot \frac{1}{2} \right) = -\frac{1}{4}$$

$$i=3 \quad x_3^{(1)} = \frac{1}{a_{33}} \left(b_3 - (a_{31} x_1^{(1)} + a_{32} x_2^{(1)}) - 0 \right)$$

$$= \frac{1}{4} \left(2 - \underbrace{(0 + 2 \cdot (-\frac{1}{4}))}_{=-\frac{1}{2}} \right) = \frac{1}{4} \left(2 + \frac{1}{2} \right) = \frac{5}{8} = \underline{\underline{0,625}}$$

$$\Rightarrow \underline{\underline{x^{(1)} = \left(\frac{1}{2}, -\frac{1}{4}, \frac{5}{8} \right)}}$$

$$\underline{k=1} \quad x_1^{(2)} = \frac{1}{4} \left(2 - 0 - a_{12} x_2^{(1)} - a_{13} x_3^{(1)} \right)$$

$$= \frac{1}{4} \left(2 - \underbrace{2 \cdot (-\frac{1}{4})}_{=-\frac{1}{2}} - 0 \right) = \frac{1}{4} \left(2 + \frac{1}{2} \right) = \underline{\underline{\frac{5}{8}}}$$

$$x_2^{(2)} = \frac{1}{4} \left(0 - a_{21} x_1^{(2)} - a_{23} x_3^{(1)} \right) = \frac{1}{4} \left(0 - 2 \cdot \left(\frac{5}{8} \right) - 4 \cdot \frac{5}{8} \right)$$
$$= \underline{\underline{-\frac{5}{2}}}$$

Klausurvorbereitung A1 -2-

$$x_3 = \begin{pmatrix} 0,8125 \\ -0,8125 \\ 0,90625 \end{pmatrix}, \quad x_4 = \begin{pmatrix} 0,90625 \\ -0,90625 \\ 0,953125 \end{pmatrix}$$

b) $A = D - E - F$; $B_{GS} = (D - E)^{-1} F$

$$A = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 4 & 2 \\ 0 & 2 & 4 \end{pmatrix} \Rightarrow D = \begin{pmatrix} 4 & & \\ & 4 & \\ & & 4 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 0 \\ 0 & -2 & 0 \end{pmatrix},$$

$$F = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow D - E = \begin{pmatrix} 4 & 0 & 0 \\ 2 & 4 & 0 \\ 0 & 2 & 4 \end{pmatrix}$$

$$\begin{array}{ccc|ccc} 4 & 0 & 0 & 1 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 & 0 & 1 \end{array} \Leftrightarrow \begin{array}{ccc|ccc} 1 & 0 & 0 & 1/4 & 0 & 0 \\ 0 & 4 & 0 & -1/2 & 1 & 0 \\ 0 & 2 & 4 & 0 & 0 & 1 \end{array} \left. \begin{array}{l} \cdot (-1/2) \\ + \end{array} \right\}$$

$$\Leftrightarrow \begin{pmatrix} 1 & 0 & 0 & 1/4 & 0 & 0 \\ 0 & 4 & 0 & -1/2 & 1 & 0 \\ 0 & 0 & 4 & 1/4 & -1/2 & 1 \end{pmatrix} \Rightarrow (D - E)^{-1} = \begin{pmatrix} 1/4 & 0 & 0 \\ -1/8 & 1/4 & 0 \\ 1/16 & -1/8 & 1/4 \end{pmatrix}$$

$$B_{GS} = \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 1/4 & -1/2 & 1 \end{pmatrix} \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 0 & -2 & 0 \\ 0 & 1 & -2 \\ 0 & -1/2 & 1 \end{pmatrix}$$

$$\Rightarrow \|B_{GS}\| = \frac{1 + |-2|}{4} = \frac{3}{4} \quad (\text{Zeilensummennorm})$$

Klausurvorbereitung A1

-3-

wg. $S(B_{GS}) \leq \|B_{GS}\|_{\infty} = \frac{3}{4}$ konvergiert das GS-Verfahren,
und wg. $\|B_{GS}\| < 1$ ist die Konvergenz in der ∞ -Norm monoton.

a) $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}, b = \begin{pmatrix} 203 \\ 102 \end{pmatrix}$

Jacobi:
 $x_0 = (0, 0)^T$

GS
 $x_0 = (0, 0)^T$

$$x_1 = (101,5; 51)^T$$

$$x_1 = (101,5; 0,25)^T$$

$$x_2 = (25,0; 0,25)^T$$

$$x_2 = (101,125; 0,4375)^T$$

$$x_3 = (101,125; 38,5)^T$$

$$x_3 = (100,84375; 0,578125)^T$$

$$x_4 = (43,75; 0,4375)^T$$

$$x_4 = (100,6328125; 0,68359375)^T$$

$$x_5 = (100,84375; 28,125)^T$$

$$x_5 = (100,4746094; 0,762695...) ^T$$

b) Jacobi-Verfahren: $B_J = D^{-1}(E+F)$ mit $A = D-E-F$

$$\Rightarrow D = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, E = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, F = \begin{pmatrix} 0 & -3 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow B_J = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} \cdot \begin{pmatrix} 0 & -3 \\ -1 & 0 \end{pmatrix} = \underline{\underline{\frac{1}{2} \begin{pmatrix} 0 & -3 \\ -1 & 0 \end{pmatrix}}}$$

Gauß-Seidel-Verf.: $B_{GS} = (D-E)^{-1}F = \underbrace{\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}^{-1}}_H F$

$$\left[\begin{array}{cc|cc} 2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right] \xrightarrow{1.2} \left[\begin{array}{cc|cc} 1 & 0 & 1/2 & 0 \\ 0 & 4 & -1 & 2 \end{array} \right] \Rightarrow H = \begin{pmatrix} 1/2 & 0 \\ -1/4 & 1/2 \end{pmatrix}$$

$$\Rightarrow B_{GS} = \begin{pmatrix} 1/2 & 0 \\ -1/4 & 1/2 \end{pmatrix} \begin{pmatrix} 0 & -3 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -3/2 \\ 0 & 3/4 \end{pmatrix}$$

$S(B_J)$: bestimme EW von B_J

$$P_{B_J}(\lambda) = |B_J - \lambda I| = \lambda^2 - 3/4 = 0 \Leftrightarrow \underline{\underline{\lambda = \pm \frac{\sqrt{3}}{2}}}$$

$$\Rightarrow S(B_J) = \frac{\sqrt{3}}{2}$$

$$S(B_{GS}): P_{B_{GS}}(\lambda) = -\lambda(3/4 - \lambda) \stackrel{!}{=} 0 \Leftrightarrow \lambda = 0 \vee \lambda = 3/4 \Rightarrow S(B_{GS}) = \underline{\underline{3/4}}$$

c) gem. Skript: der optimale Relaxationsparameter ω_{opt} ist:

$$\omega_{opt} = \frac{2}{2 - \lambda_1 - \lambda_n} = \frac{2}{2 - (\lambda_{min} + \lambda_{max})}$$

$$\text{hier: } \lambda_{min} = -\frac{\sqrt{3}}{2}, \lambda_{max} = \frac{\sqrt{3}}{2} \Rightarrow \underline{\underline{\omega_{opt} = 1}}.$$

Das Jacobi-Verfahren ist hier bereits optimal relaxiert.

a) Satz von Gerschgorin: $\lambda \in W$ in $\bigcup_{i=1}^n K_i$ mit

$$K_i = \{ |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \}$$

hier: $K_1 = \{ |z - 0| \leq 4 \}$

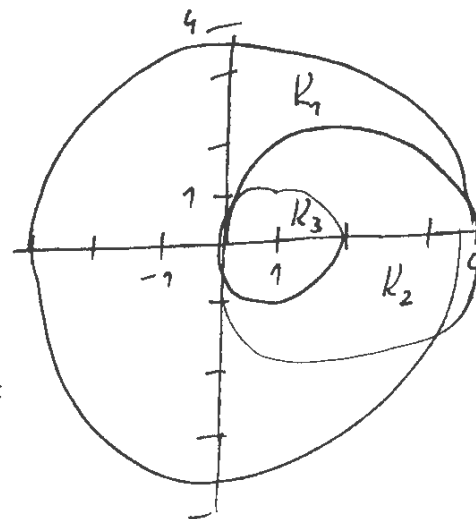
$$K_2 = \{ |z - 2| \leq 2 \}$$

$$K_3 = \{ |z - 1| \leq 1 \}$$

wg. $K_1 \supset K_2$ und $K_1 \supset K_3$:

$$\bigcup_{i=1}^n K_i = K_1$$

$$\Rightarrow \underline{\underline{|\lambda| \leq 4}}$$



b) $P_A(\lambda) = \begin{vmatrix} -\lambda & -4 & 0 \\ 0 & 2-\lambda & -2 \\ 0 & -1 & 1-\lambda \end{vmatrix} \stackrel{\text{Laplace}}{=} -\lambda \begin{vmatrix} 2-\lambda & -2 \\ -1 & 1-\lambda \end{vmatrix}$

$$= -\lambda ((2-\lambda)(1-\lambda) - 2)$$

$$= -\lambda (2 - 3\lambda + \lambda^2 - 2) = -\lambda^2 (\lambda - 3)$$

$$\Rightarrow \lambda \in W: 0, 3$$

c) $x_0 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow y_1 = Ax_0 = \begin{pmatrix} 0 & -4 & 0 \\ 0 & 2 & -2 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix}$

$$\Rightarrow x_1 = \frac{y_1}{\|y_1\|} = \frac{1}{\sqrt{21}} \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix} \approx \begin{pmatrix} -0,8728715608... \\ 0,4364357804... \\ -0,2182178902... \end{pmatrix};$$

$$\lambda_1 = \langle x_0, y_1 \rangle = \underline{\underline{2}}$$

$$x_2 = \begin{pmatrix} -0,766261028... \\ 0,574695771... \\ -0,287347885... \end{pmatrix}, \quad \lambda_2 = \langle x_1, y_2 \rangle = 2,23809523...$$

$$x_3 = \begin{pmatrix} -0,766261028... \\ 0,574695771... \\ -0,287347885... \end{pmatrix}, \quad \lambda_3 = \langle x_2, y_3 \rangle = 3,000000008$$

$$x_4 = \begin{pmatrix} " \\ " \\ " \end{pmatrix}, \quad \lambda_4 = \langle x_3, y_4 \rangle = 3,000000003$$

Klausurvorbereitung A5

a) $f(0) = 0 - 1 = -1 < 0$, aber $f(x) = \sin x - \cos x$

$$f\left(\frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) - \cos\left(\frac{\pi}{2}\right) = 1 - 0 > 0 \quad \text{und} \quad \frac{\pi}{2} \in [0, 2]$$

zws
 $\Rightarrow \exists \xi \in [0, 2]: f(\xi) = 0$

b) $a_0 = 0, b_0 = 2 \Rightarrow x_0 = 1 \Rightarrow f(x_0) \approx 0,30 > 0$

$$\Rightarrow f(x_0) \cdot f(a_0) < 0 \Rightarrow \text{NST in } [0, 1]$$

$$a_1 = a_0 = 0, b_1 = x_0 = 1 \Rightarrow x_1 = \frac{1}{2} \Rightarrow f\left(\frac{1}{2}\right) \approx -0,38 < 0$$

$$\Rightarrow f(x_1) \cdot f(b_1) < 0 \Rightarrow \text{NST in } \left[\frac{1}{2}, 1\right]$$

$$a_2 = x_1 = \frac{1}{2}, b_2 = b_1 = 1 \Rightarrow x_2 = \frac{3}{4} \Rightarrow f\left(\frac{3}{4}\right) \approx -0,05 < 0$$

$$\Rightarrow f(x_2) \cdot f(b_2) < 0 \Rightarrow \underline{\underline{\text{NST in } \left[\frac{3}{4}, 1\right]}}$$

c) $f'(x) = \cos x + \sin x, \quad x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$

$$x_0 = \frac{\frac{3}{4} + \frac{1}{4}}{2} = \underline{\underline{\frac{7}{8}}}$$

$$\Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \approx 0,7851576 \dots$$

$$x_2 = 0,785398 \dots$$

$$\underline{\underline{x_3 = 0,785398 \dots}}$$

3 Klausurvorbereitung A6

-1-

a) x_i $y_i = d_{ii}$

-1	5		
0	3	$\rightarrow \frac{3-5}{0-(-1)} = -2$	
1	5	$\frac{5-3}{1-0} = 2$	$\rightarrow \frac{2-(-2)}{1-(-1)} = 2$
3	21	$\frac{21-5}{3-1} = 8$	$\frac{8-2}{3-0} = 2 \rightarrow 0$

$$d_{mm} = \frac{d_{m,m+1} - d_{m-1,m}}{x_m - x_{m-1}}$$

$$\Rightarrow \underline{\underline{p = 5 - 2(t+1) + 2t(t+1) + 0(t+1)t(t-1)}}$$

b)

$$\text{LGS: } \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 9 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \underbrace{\begin{pmatrix} 5 \\ 3 \\ 5 \\ 21 \end{pmatrix}}_{=C}$$

$$A^T A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 9 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 9 \end{pmatrix} = \begin{pmatrix} 4 & 11 \\ 11 & 83 \end{pmatrix};$$

$$A^T b = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 9 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \\ 5 \\ 21 \end{pmatrix} = \begin{pmatrix} 34 \\ 199 \end{pmatrix}$$

$$\text{Löse } A^T A \cdot \begin{pmatrix} a \\ b \end{pmatrix} = A^T C$$

$$\begin{array}{cc|cc} 4 & 11 & 34 & \cdot 11 \\ 11 & 83 & 199 & \cdot 4 \end{array}$$

$$\Leftrightarrow \begin{array}{cc|cc} 44 & 121 & 374 & \\ 44 & 332 & 796 & \end{array} \Leftrightarrow \begin{array}{cc|cc} 44 & 121 & 374 & \\ 0 & 211 & 422 & \end{array}$$

Klausurvorbereitung, Ab -2-

$$\Leftrightarrow \begin{array}{cc|c} 4 & 11 & 34 \\ 0 & 1 & 2 \end{array} \xrightarrow[-11]{+} \begin{array}{cc|c} 4 & 0 & 12 \\ 0 & 1 & 2 \end{array} \Rightarrow \underline{\underline{a=3, b=2}}$$

$$\Rightarrow \underline{\underline{p(x) = 3 + 2x^2}}$$

Klausurvorbereitung

A7

$$I = \int_0^4 \frac{x}{1+x} dx$$

$$T_1 = 4 \cdot \frac{0 + \frac{4}{1+4}}{2} = 2 \cdot \frac{4}{5} = \underline{\underline{\frac{8}{5}}}$$

$$T_2 = \frac{h}{2} (f(0) + 2f(2) + f(4)) \quad \text{mit } h = \frac{4}{2} = \frac{b-a}{n}$$

$$= 1 \left(0 + 2 \cdot \frac{2}{3} + \frac{4}{5} \right) = 1 \left(\frac{4}{3} + \frac{4}{5} \right) = 1 \cdot \frac{20+12}{15} = \underline{\underline{\frac{32}{15}}}$$

$$T_4 = \frac{1}{2} (f(0) + 2f(1) + 2f(2) + 2f(3) + f(4))$$

$$= \left(0 + 2 \cdot \frac{1}{2} + 2 \cdot \frac{2}{3} + 2 \cdot \frac{3}{4} + \frac{4}{5} \right) \cdot \frac{1}{2}$$

$$= \frac{1}{2} \left(1 + \frac{20}{15} + \frac{6}{4} + \frac{12}{15} \right) = \frac{60 + 80 + 90 + 48}{2 \cdot 60} = \underline{\underline{\frac{138}{60}}}$$

Klausur vorbereitung A8

- 1 -

a) Es gilt $\int \frac{1}{1+x^2} dx = \arctan x$, also

$$\int_0^1 \frac{1}{1+x^2} dx = \arctan(1) - \arctan(0) = \frac{\pi}{4} - 0 = \underline{\underline{\frac{\pi}{4}}}$$

b) $\frac{1}{1+x^2}$ ist streng monoton fallend auf $[0, 1]$, so dass man die Obersumme O_n durch Auswertung von f am linken, die Untersumme U_n durch Ausw. von f am rechten Teilintervall erhält.

$$\Rightarrow O_n = \frac{1}{n} \sum_{i=0}^{n-1} f\left(\frac{i}{n}\right) ; \quad U_n = \frac{1}{n} \sum_{i=1}^n f\left(\frac{i}{n}\right)$$

$$D_n = O_n - U_n = \frac{1}{n} (f(0) - f(1)) = 0,05 \quad (*)$$

c) $O_n = \frac{1}{10} (f(0) + f(\frac{1}{10}) + \dots + f(1))$

$$= \underline{\underline{0,803381\dots}}$$

$$U_n = O_n - D_n = 0,803381 - 0,05 = \underline{\underline{0,753381\dots}}$$

d) $D_n < \frac{\pi}{4} - 0,785 = \frac{1}{n} \cdot (1 - \frac{1}{2}) = \frac{1}{2n}$

$$\Leftrightarrow n > \frac{\frac{1}{2}}{\frac{\pi}{4} - 0,785} \approx 1255,7$$

Klausurvorbereitung A8 -2-

$$\begin{aligned}
 e) \quad S &= \frac{1}{6} (f(0) + 4f(\frac{1}{2}) + f(1)) = 0,7 \\
 &= \frac{1}{6} \left(1 + 4 \cdot \frac{1}{1 + \frac{1}{4}} + \frac{1}{2} \right) = \frac{1}{6} \left(1 + 4 \cdot \frac{4}{5} + \frac{1}{2} \right) \\
 &= \frac{1}{6} \frac{10 + 32 + 5}{10} = \frac{47}{60} = 0,78\bar{3}
 \end{aligned}$$

$$e_{abs} = |S - \frac{\pi}{4}| \approx \underline{\underline{2,06 \cdot 10^{-3}}}$$

f) Streifenzahl 10:

$$\begin{aligned}
 S_{10} &= \frac{1-0}{10 \cdot 3} \left(f(0) + 4f(\frac{1}{10}) + 2f(\frac{2}{10}) + 4f(\frac{3}{10}) + \dots + f(1) \right) \\
 &= 0,78539815348\dots
 \end{aligned}$$

$$e_{abs} = |S_{10} - \frac{\pi}{4}| \approx 9,91 \cdot 10^{-9}$$

g) i) Umrechnung des Gewichts: $\tilde{\beta}_i = \frac{b-a}{2} \beta_i = \frac{1}{2} \beta_i = \underline{\underline{\frac{1}{2}}}$

ii) Umrechnung der Stützstellen $\tilde{x}_i = \frac{b-a}{2} x_i + \frac{b+a}{2}$

$$= \frac{1}{2} x_i + \frac{1}{2}$$

$$g_1: x_i = \pm \frac{1}{\sqrt{3}} \Rightarrow \tilde{x}_1 = \frac{1}{2} \left(\frac{-1}{\sqrt{3}} \right) + \frac{1}{2} = \frac{1}{2} - \frac{1}{2\sqrt{3}} = \underline{\underline{\frac{1}{2} \left(1 - \frac{1}{\sqrt{3}} \right)}}$$

$$\underline{\underline{\tilde{x}_2 = \frac{1}{2} \left(1 + \frac{1}{\sqrt{3}} \right)}}$$

$$\Rightarrow g_1 = \tilde{\beta}_1 f(\tilde{x}_1) + \tilde{\beta}_2 f(\tilde{x}_2) \approx \underline{\underline{0,786885\dots}}$$

$$\Rightarrow e_0 = 1,49 \cdot 10^{-3}$$

Klausurvorbereitung AG

-1-

$$a) \quad n=2 \Rightarrow 2n-1=3$$

$$\text{rechte Seite: } \left. \begin{aligned} \int_a^b x^0 dx &= \int_{-1}^1 1 dx = 2 \\ \int_{-1}^1 x dx &= 0; \quad \int_{-1}^1 x^2 dx = \frac{2}{3}; \quad \int_{-1}^1 x^3 dx = 0 \end{aligned} \right\} b = \begin{pmatrix} 2 \\ 0 \\ \frac{2}{3} \\ 0 \end{pmatrix}$$

$$\Rightarrow \text{Gleichungen: } w_1 x_1^0 + w_2 x_2^0 = 2$$

$$w_1 x_1 + w_2 x_2 = 0$$

$$w_1 x_1^2 + w_2 x_2^2 = \frac{2}{3}$$

$$w_1 x_1^3 + w_2 x_2^3 = 0$$

Einsetzen von w_1, w_2, x_1 und x_2 zeigt, daß alle 4 Gl. erfüllt sind.

b) Aufgrund des Intervalls $[-1, 1]$ entfällt die Umrechnung der β_i bzw. x_i .

$$\Rightarrow g_1(f) = \beta_1 f(x_1) + \beta_2 f(x_2) = \beta_1 e^{x_1} + \beta_2 e^{x_2} \approx 2,342696...$$

$$\Rightarrow e_{g_1} = \left| g_1(f) - \int_{-1}^1 e^x dx \right| \approx \underline{\underline{7,706 \cdot 10^{-3}}}$$

$$c) \quad \text{Trapezregel: } T = \frac{2}{2} (f(-1) + f(1)) = e + \frac{1}{e} \approx 3,08616$$

$$\text{exakt: } e - \frac{1}{e} \Rightarrow \text{Fehler: } \frac{2}{e} \approx 0,73...$$

$$\text{Simpson: } S = \frac{2}{6} (f(-1) + 4f(0) + f(1)) = \frac{1}{3} \left(e + 4 + \frac{1}{e} \right)$$

Klausurvorbereitung AG -2-

$$S = 2,362053... \Rightarrow \text{Fehler: } \approx 1,16 \cdot 10^{-2}$$

$$d) \quad e_a \leq \frac{M(b-a)^3}{12m^2} \stackrel{!}{\leq} \underbrace{7,706 \cdot 10^{-3}}_{:= e_g} \quad (*)$$

$$M = \max_{[-1,1]} |f''(x)| = \max_{[-1,1]} e^x = e$$

$$\frac{e \cdot 8}{12m^2} \leq e_g \quad \Leftrightarrow \quad \frac{8e}{e_g} \leq 12m^2$$

$$\Leftrightarrow \frac{8}{12} \frac{e}{e_g} \leq m^2 \Rightarrow m > \sqrt{\frac{2}{3} \frac{e}{e_g}} \approx 15,33..$$

$$\Rightarrow \underline{\underline{m=16}}$$

$$e) \quad T_{16} = \frac{2}{16} \left(f(-1) + 2f\left(-\frac{7}{8}\right) + 2f\left(-\frac{6}{8}\right) + \dots + 2f\left(\frac{7}{8}\right) + f(1) \right)$$

$$\approx \underline{\underline{2,353462...}}$$

Vektor mit den Funktionswerten

0,367879...	1,13314...
0,416820...	1,28402...
0,472366...	1,45488...
0,535261...	1,64872...
0,606530...	1,86824...
0,687288...	2,11700...
0,778800...	2,38887...
0,882486...	2,71828...
1	

Klausurvorbereitung AG -3

$$f) S_{16} = \frac{2}{16} \left(f(-1) + 4 f\left(-\frac{7}{8}\right) + 2 f\left(-\frac{6}{8}\right) + 4 f\left(-\frac{5}{8}\right) + \dots + 2 f\left(\frac{7}{8}\right) + f(1) \right)$$

$$\approx 2,35045568 \quad ; \quad \underline{\underline{\text{Fehler: } 3,182 \cdot 10^{-6}}}$$